PART 1

MODELLING AND SYNTHESIS

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1

INTRODUCING MODELLING AND SYNTHESIS FOR STRUCTURAL INTEGRITY

With them the seeds of Wisdom did I sow,
And with mine own hands wrought to make it grow; Edward Fitzgerald, The Rubaiyat of Omar Khayyám, 1889

This is a book about the basic elements of engineering design. We begin with an emphasis on structural integrity, and later broaden our concern to various philosophical and practical issues. The book is addressed to engineering undergraduates who are being exposed to engineering design thinking and reasoning for the first time in their courses. Many texts on engineering design of machine elements treat structural integrity from a purely analytical point of view. Readers of such texts are rarely offered opportunities to apply the process of design synthesis to *real-life* engineering problems. Engineering texts present encapsulated experiences with design problems pared down to a manageable scale. *Real-life* engineering is full of uncertainties and risks, impossible to replicate effectively in the formalised medium of a textbook. Nevertheless, first-time students of engineering need to be introduced to the myriad choices and decisions that designers must face.

Our focus in this first part of the text is on design for structural integrity. For the purposes of our study, we define structural integrity as the capacity of engineering components to withstand service loads, effectively and efficiently, during their service life. This definition involves several concepts that need further elucidation.

Engineering component in this context means any engineering structure, which may be constructed from several interconnected elements into a *single entity*. Examples are pressure vessels, bicycle frames, flywheels, springs, shafts of electric motors, airframes and the frames of motor vehicles or buildings;

Effectively withstanding loads is defined as the capacity to accept service loads without exceeding either the specified maximum stress, specified maximum deflection, or both of these specifications. Examples are: 140 MP a maximum shear stress allowed in certain qualified welding materials, or maximum deflection of 1/300 the span in architectural beams;

Service loads are those loads, specified or unspecified, that the designer considers as credible to be imposed on the component during its service life. In the context of structural integrity, a *service load* will usually be specified as a *force* (N), a *moment* (Nm), or a *pressure* (Pa). More generally, *design loads* can be thought of as *thermal* (kW/m^2), *electrical* (A/m^2; kW), *information-processing* (decisions/hr; bits), or even in terms of such concepts as *traffic intensity* (e.g. passengers/hr).

The mature description of service loads is probably the aspect of component design requiring the greatest creative input by the designer, in terms of engineering wisdom and judgement. Often the service loads are unknown and need to be estimated before design can begin. In the case of critical design, such as airframes, considerable in-service test data is used to enable suitable service load estimates. For critical components, and those subject to fatigue failure, (e.g. airframes and motor vehicle components) these estimates are almost invariably followed up by field testing of prototypes. We explore service loads a little further in section 1.3, *Structural distillation*.

Efficiently, in this context, means either *at least cost* or *at least mass*. Cost-limited designs, the most common case, must consider all aspects of cost, including material, manufacture, maintenance, and most importantly, design

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costs. In general, for relatively simple components, the cost of manufacture is of the same order of magnitude as the material cost. In contrast to this, design time is expensive and needs to be spent wisely. Hence, the wise, or experienced designer will not spend expensive design time to eliminate a few extra grams of material from a relatively simple and cheap component. Quite often, for simple components, *one that does the job* is likely to be most efficient. Mass-limited designs require considerable analysis. With complex shapes, finite element analysis (a computer-based stress analysis procedure) offers opportunities for mass optimisation[1.1].

For the purposes of design, once the service loads have been established, the mechanical analysis of a component requires mathematical modelling of the component structure. The model allows us to translate loads into stresses or deflections. Naturally, the more closely the mathematical model represents reality, the more accurately we can identify stresses or deflections. As a precursor to analysis, we must make absolutely clear that the results of our analyses reflect only the behaviour of our model, and not the behaviour of the *real* component. Even experienced designers sometimes overlook this consideration. The spectacular failure of the *Tacoma Narrows* suspension bridge is an example of the complacency of structural engineers with models. As Professor Henry Petroski notes (Petroski, 1994):

"In the half-century following the accomplishment of the Brooklyn Bridge, suspension bridge design evolved in a climate of success and selective historical amnesia... to the Tacoma Narrows Bridge. Now, a half-century after that bridge's colossal collapse, there is reason to be concerned that the design of newer bridge types will be carried out without regard to apparent cycles of success and failure."

The Tacoma Narrows Bridge was of lighter design than previously built bridges, since it was designed to carry vehicular, rather than rail traffic. Wind loads on previously designed suspension bridges were modelled as causing static lateral deflections only. The lighter deck structure of the Tacoma bridge was susceptible to severe torsional vibrations, a loading clearly overlooked in the design, and this eventually caused its failure (Farquharson, 1949)[1.2].

Another famous case of improper modelling is the recorded tragic disasters suffered by the Comet aircraft of the DeHavilland company in the early 1950s. Although controversial at the time, it is now generally agreed that the aircraft failures were due to stress concentrations near the sharp corners of some rectangular windows in the fuselage. It seems clear that the Comet designers were unaware of the significant variability of fatigue data, and failed to take this into account in their design (Hewat and Waterton, 1956).

On a grand scale, we accepted Sir Isaac Newton's models of the physics of motion universally for nearly 400 years. In fact, they were, and still are, referred to as *Newton's Laws*. Einstein's hypotheses on relativity forced a paradigm shift in physics thinking. We no longer regard the *laws* of physics as absolute truths, but merely as conjectures, or working hypotheses, to be used until refuted by physical experiment (Popper, 1972).

In Section 1.1 we identify some characteristic features of engineering failures. We start by offering four propositions about failures, and we describe some simple engineering tools to be used in design for structural integrity (*failure avoidance*). We also offer some spectacular examples of failures, characterised by their social, economic or environmental impact.

The key elements of design for structural integrity are:

- service load estimation; and
- structural modelling.

Both of these elements require sound engineering judgement. We need to establish a base reference frame for our capability in making such judgements. In Section 1.2 we explore the notion of *engineering estimation*. We also set the reference frame for what we expect first-time users of this text to know, in order to make effective use of the remaining chapters.

Material properties and material selection are closely interwoven with design for structural integrity. Chapter 2 introduces material properties, formal definitions of failure and failure predictors to be used in design for structural integrity.

In Chapters 3 and 4 we offer simple mathematical models for several generic engineering components: *columns*, *shafts*, *pressure vessels*, *springs* , as well as *bolted joints*, *pinned connections* and *welded joints*. These components, and their respective mathematical models, represent the basic building blocks of our approach to modelling the failure–behaviour of more complex engineering structures.

1.1 Minimum mass with imposed stress or deflection constraints.

1.2 Quoted in Petroski (1994)

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We propose that most engineering structures may be treated as a combination of several cooperating, much simpler, engineering structural elements. Moreover, if we understand how these simpler elements behave under loads, and we can recognise the underlying, simpler engineering elements which constitute a given *practical* structure, we are exercising *structural distillation* and *design thinking*.

This simplistic approach to structural integrity is clearly applicable to practical design situations where a relatively conservative or *safe* design is required. For designs of complex structures, exposed to limit loads (or beyond in some cases, such as racing cars, or racing yachts), we advocate the application of more sophisticated techniques of structural analysis. Nevertheless, even the more subtle problems become easier to understand and to explore, once we have a good basis for understanding how simpler structures behave. In Section 1.3 we introduce *structural distillation*, and we investigate its application to a wide range of common engineering components.

1.1 Structural integrity and the nature of failure

Proposition 1: Engineers are inherently concerned with failure and our vision of success is to develop modelling tools to avoid it. Moreover, by studying failures we develop clear ideas about causal relationships in complex *real-life* engineering situations, often too difficult to model completely realistically for structural analysis.

Proposition 2: Engineering failures may be categorised as technical, operational or unpredictable:

- technical failures are most commonly due to insufficient information about the nature of the structure, its material, its loading, or its operating conditions;
- operational failures are most commonly due to improper operating practices or conditions;
- unpredictable failures are most commonly the result of special circumstance or *acts of God*.

Proposition 3: Incentives to avoid engineering failures are related to *failure intensity* – the degree of seriousness of the failure. *Failure intensity* is measured by the economic, environmental or social impact caused by the failure.

Proposition 4: Failure is essentially related to risk: given extreme conditions, all structures or systems can fail. The engineer's task is to:

- generate design specifications that best meet the required operating conditions of the structure or system within acceptable levels of risk;
- identify the limits of *know–how* associated with the structure or system and assign *factors of ignorance* (commonly referred to as *factors of safety*) to cope with these limits, also within acceptable levels of risk — this brings with it the notion of *worst credible accident*.

1.1.1 Some spectacular structural failures

In what follows we present some spectacular structural failures, and examine the causes and outcomes associated with them.

(a) Oil tank collapse in the field

Figure 1.1 shows two of six collapsed oil tanks at a West African installation. The tanks were 15.5m diameter and 16.5m in height. At the time of erection, the tops were being assembled when a severe storm struck. The wind blowing over the tops exerted both direct pressure on the outside of the tanks as well as causing negative pressure inside the tank on the walls facing into the wind. The loss incurred was more than $US 1 million.

According to the investigators, there were several contributing factors to this loss. All six tanks had been erected to their full height before construction of the top covers commenced, and the tanks had not been reinforced internally to support them in a storm. Furthermore, the construction took place at a time known to be the worst storm season in that part of the world.

While no lives were lost nor any human injuries incurred, the financial loss was substantial.

(b) Collapsed hoarding in Dubai

Figure 1.2 shows a collapsed hoarding around a clock-tower being built in Dubai. The hoarding was erected to shield the site from view during construction. In a storm, resulting in maximum wind speeds of 62 km/h, the hoarding collapsed

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Figure 1.1 Collapsed oil tanks in West Africa (2 of 6 similarly damaged tanks shown): After Schaden Spiegel 2/94[1.3]

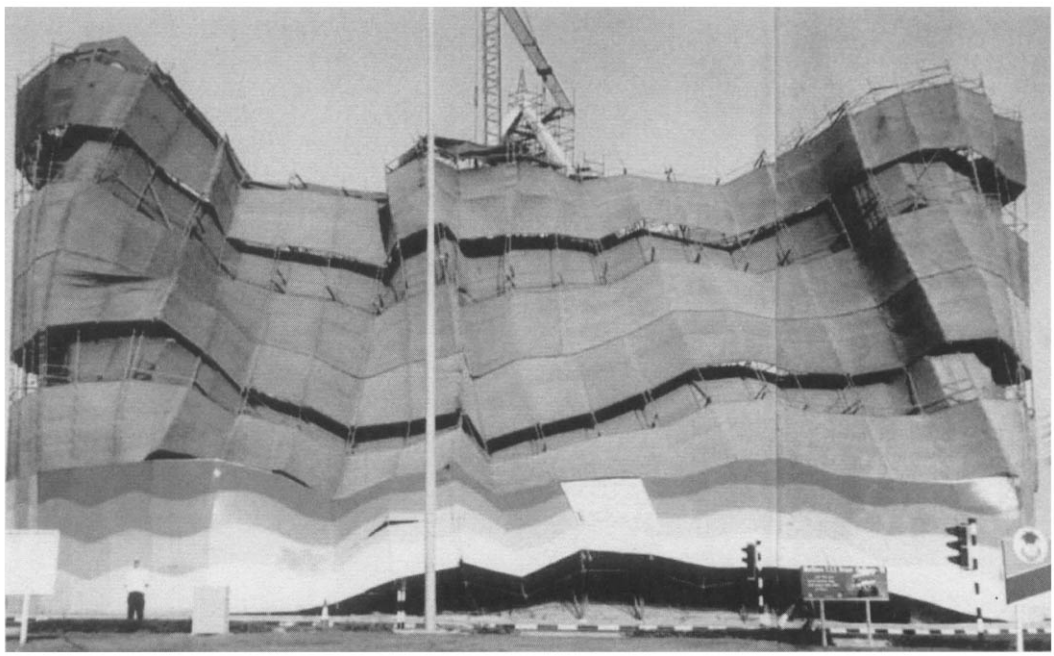


Figure 1.2 Collapsed hoarding in Dubai: After Schaden Spiegel 2/94

1.3 Schaden Spiegel (*'reflections of failure'*) is a publication of Munich Reinsurance Company, Munich, Germany

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and caused damage to the building and a site crane. The total damage was $US 400,000.

Investigators found that the hoarding was allegedly designed to withstand wind speeds of up to 115 km/h. However a design error resulted in the failure. The contractor had to bear the cost of the damage, since design errors were excluded from the insurance policy covering the works.

(c) Terrorist bomb attack on city

On April 24 1993, a terrorist bomb was used to devastate a whole section of Bishopsgate, a London suburb. One person was killed, and 40 were injured. The major loss in this event was due to the resulting fires. The damage bill came to about $US 1 billion (Figure 1.3).

While we needn't speculate on the cause, it is surprising how little damage the larger structures suffered. The anti-terror experts claim that, due to its unpredictability, this type of accident can not be designed against. However, substantial loss minimisation is possible.

Some suggested measures are:

- reduction in size of fire areas;
- installing supply facilities and safety apparatus in different fire areas; e.g. emergency power generation on roofs;
- installing safety shutoffs in electrical equipment.

(d) Kobe-Osaka highway collapse, 1995

On January 17 1995, the Kobe–Osaka highway suffered extensive damage during an earthquake. The quake lasted 40 seconds and was measured at Richter 7.2. Experts claim that the highway failure was initiated by shear failures of the piers. The total loss is estimated in billions of dollars (Figure 1.4).



Figure 1.3 Bomb devastated city, Bishopsgate, London: After Schaden Spiegel 2/94

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Clearly, this disaster was an unexpected *freak event*, and the highway designers could not regard such an event as a *credible accident* for design purposes. However, special design measures can be, and are, taken in earthquake-prone locations.

1.1.2 Some spectacular mechanical failures

This is a convenient point in our discussions of failures to make a general observation, based on the data from Munich Reinsurance, one of the largest and most experienced industrial insurers in the world today.

In the parlance of insurance assessment, disasters caused by unpredictable natural events are referred to as *acts of God*. Until 1993, the greatest loss in any one disaster was that due to hurricane Andrew, affecting the east coast of USA and the Bahamas. Wind speeds reached 280 km/hr and 44 deaths were recorded with an estimated property loss of $US 30 billion. The following is a more recent report: (from *The Age*, Melbourne, 10 February 1999):

"Hurricanes, floods and other natural disasters cost more than £53 billion ($A135 billion) in damage worldwide last year, an increase of almost 50% on 1997, according to figures from the world's largest insurance company. The rise in the scale and scope of natural disasters, linked to global warming and reflected in a report produced by Munich Re, has alarmed underwriters so much that some are considering making parts of the world uninsurable."

Of all structural or engineering disasters around the world, those due to earthquake and windstorm represent 96% of total losses. In general, a very large proportion of industrial loss is incurred due to fire damage. Consequently, while mechanical failures may be embarrassing, occasionally very costly, and in some cases result in tragedy (see for example Eddy Potter and Page, 1976), on the scales of insured industrial loss, they are a *rare event*. This doesn't mean to imply that they happen rarely. On the contrary, engineering components fail with



Figure 1.4 Collapsed Kobe-Osaka highway

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monotonous regularity. However, in most cases, designers are aware of the limitations of engineering components and systems. Failures elicit planned responses, and damage and injury are minimised. In some critical designs, and in design against fatigue, mechanical failures are often used as a design tool to establish component life.

Some very special engineering components are *designed to fail* in service. On critical pressure vessels a *bursting-disc* is used to limit the effects of overpressure to a specially designed part of the system. In some machinery, expensive parts are protected by specially designed, cheap, easily replaced, sacrificial components. These types of *mechanical fuses* minimise damage and injury. What engineering designers fear most is unexpected and un-planned-for failures. Some are described in the next section. In general, if properly investigated and documented, they represent valuable case studies for future designs.

(a) Failure of 600 MW turboset in a steam power generating plant

Turbogenerators are common types of engineering systems used to convert thermal energy, usually from steam or gas, into mechanical and thence electrical energy. The subject of this failure example was steam driven by a two stage turbine set.

Following a major overhaul, carried out after 25,000 successful service hours, the 600 MW turbogenerator set was being checked for overspeed protection. After successfully checking the electrical overspeed protection, at 108% of rated speed, the machine speed was reduced and then increased to check the mechanical overspeed protection, set at 110% of the rated speed. Just prior to reaching this overspeed, the whole set *exploded* causing 85% destruction of the turbine and generator. The loss was estimated at $US 40 million.

The generator shaft was ruptured over half its length (seen in Figure 1.5), and the entire train of

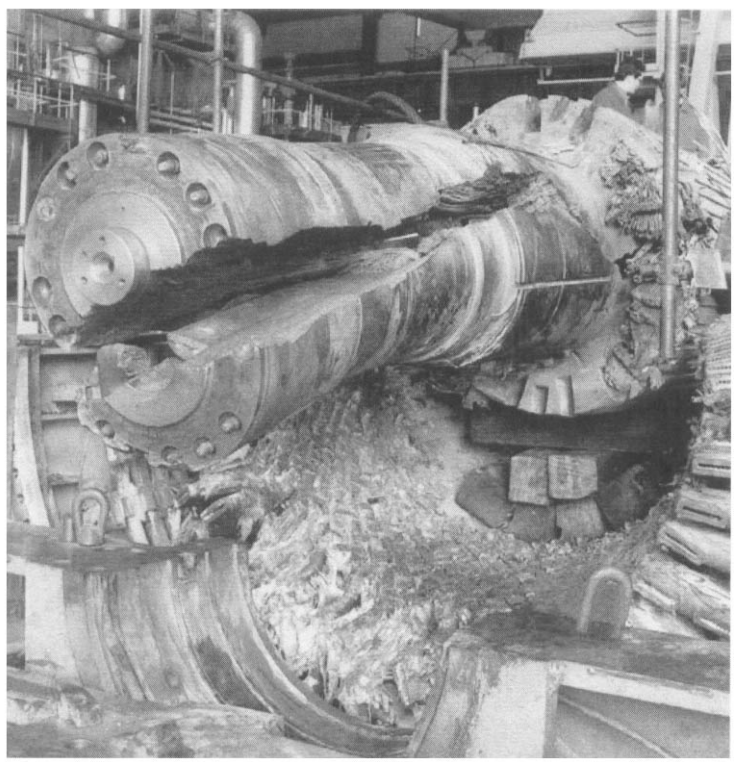


Figure 1.5 Failure of the main shaft of a 600 MW turbogenerator set: After Schaden Spiegel 1/82

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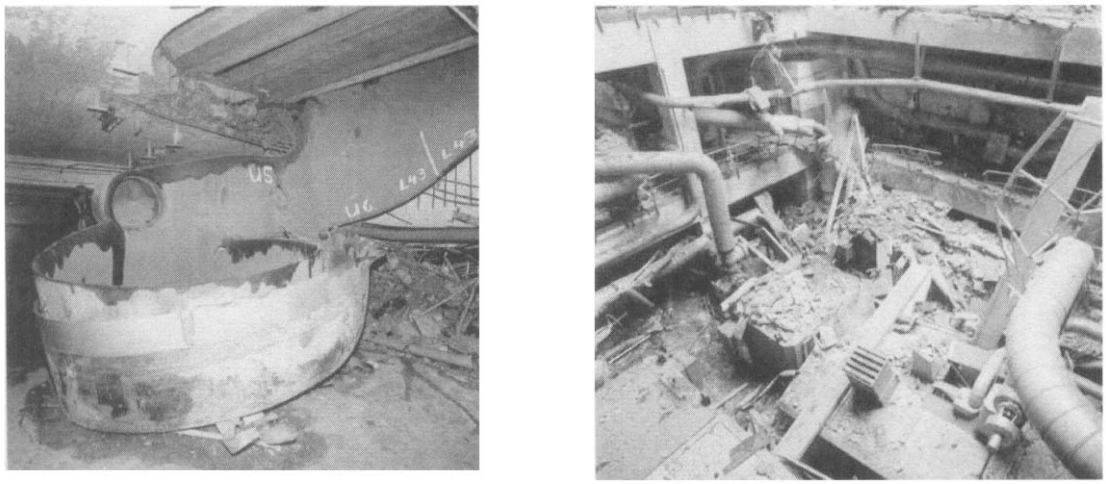


Figure 1.6 Burst high pressure preheater and devastated plant room: After Schaden Spiegel 1/82

shafts was broken in 11 places. One rotating part of the generating equipment, weighing about three tonne, was hurled into the machine hall and hit the girder of an overhead travelling crane.

The insurer's experts speculated on the cause, but they could not find any substantive reason for the failure. No material defects were found and all shaft failures were attributed to exceeding the permissible bending stresses. Subsequent speculation by rotodynamics experts conjectured that a bearing or rotor instability might have caused this type of failure.

(b) Explosion of high pressure preheater

High pressure feedwater preheaters are commonly used in steam turbine plants as a means of usefully employing exhaust steam, from the high pressure stage of the turbine. The subject of this failure was a 2m diameter, 10m high welded steel tank, with a 22mm outer jacket, operating at 300°C and 33 bars pressure (3.34MPa). Due to an emergency shutdown, the pressure in the preheater rose to 37.5 bars (3.80MPa) and the outer jacket burst, resulting in the explosion of the tank and severe damage to the plant room. The loss incurred was approximately $US 3 million.

The burst vessel and the devastated plant room are shown in Figure 1.6. Initial investigation of this failure focused on the vessel material. There was some concern that other similar vessels in this plant could suffer similar failures from what appeared

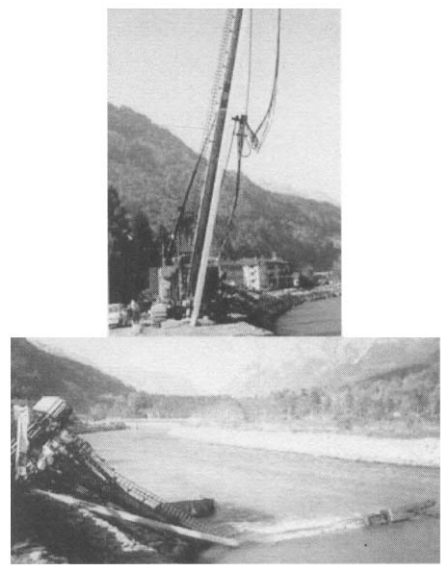


Figure 1.7 Collapsed piledriver: After Schaden Spiegel 2/96

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to be a relatively minor pressure rise, well below the safe design limit. Ultimately, however, the failure was attributed to faulty welding of this particular vessel. The faulty weld was a longitudinal weld, with a manufacturing fault, that remained undetected during X-ray inspection. During 11,500 hours of operation the faults in the weld developed further, requiring only the small pressure rise to initiate the failure.

(c) Piledriver collapse

Piledrivers are essentially large mechanical hammers, used to drive large pylons (piles) into the ground during the preparatory stages of major construction works. In some pile drivers the hammer has a centrifugal out-of-balance mass driven to provide a vibrating load for hammering. The angle of the hammer guides needs to be adjustable to allow shifting of the centre of gravity of the unit. This adjustment is particularly important when operating on uneven ground, where a very small *lean* (out of vertical) of the hammer guides can result in substantial overturning moments acting on the unit. A 30m high, 140 tonne, piledriver with a vibrating hammer was being used for earthworks near the edge of a river, when the unit, driven on a crawler tractor, slowly toppled over into the river (Figure 1.7). The recovery and repair of the pile driver was estimated at approximately $US 1 million.

Investigation of this failure found that a jammed hydraulic valve prevented successful hydraulic control of the 30m high hammer guides. Once the unit commenced to lean over in the uneven soil of the river bank, the hydraulics could not correct the lean by appropriate shift of the centre of gravity and the unit toppled over into the river.

(d) Broken Liberty ships

In September 1941, the *Patrick Henry*, a rugged 10,490 tonne merchant ship, was launched in the US. This was the first of 2,210 Ec-2 class ships that became known as the *Liberty* ships. War time pressures of production schedules demanded that these ships should be produced in record time. One such Liberty ship was delivered fourteen days after laying the keel in the shipyard.

These ships were of low alloy steel, fully welded, construction. In earlier, riveted hull ships, structural cracks would be halted by the presence of rivet holes in the material.

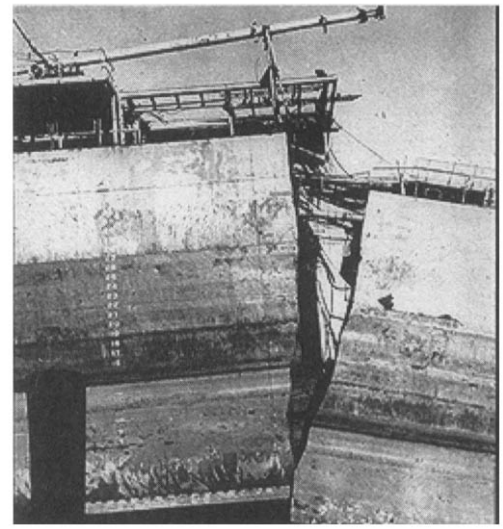


Figure 1.8 Broken Liberty ship

In a welded construction, any crack, once initiated will continue to propagate through the structure.

Some of the Liberty ships were scheduled to operate in very cold climates. The combination of low temperature embrittlement (lowered fracture toughness in low temperatures) and the unimpeded propagations of cracks through the structure resulted in some spectacular failures in these ships (Figure 1.8).

(e) Turbine blade failure

Figure 1.9 shows a type of turbine blade connection between the disc and blading, commonly referred to by the geometry-inspired title of *fir-tree* connection. A photo elastic model of a similar *fir-tree* connector under load is also shown in the figure. Photo-elastic models are made of a photo-polymer, which can cause light to split into two components when under stress (double refracting). When polarised light is passed through a photopolymer under stress, the stresses show up as interference patterns. These patterns can display the stress in the material, with closely spaced fringe lines indicating large stress gradients.

From Figure 1.9 we can infer that the stresses in the model are high in the root region of the *fir-tree* connector and also around the *teeth* of the *fir-tree*. Figure 1.10 shows a failed turbine disc, with the blades torn out of the disc. Failures of this type can result in spectacular and costly damage. In one

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case, reported in *Schaden Spiegel 1/82*, a blade disc failure in a 120MW turbine caused a loss of $US 5 million. Several fragments cut through the turbine casing, damaging it beyond repair. One fragment impacted a 100,000kVA transformer causing oil to leak out and ignite. The major damage was caused by the ensuing fire.

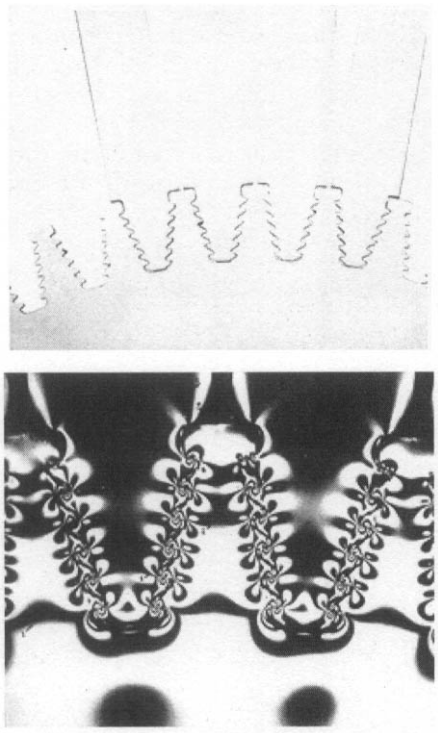


Figure 1.9 Fir-tree connection between turbine disc and blading and corresponding photoelastic model

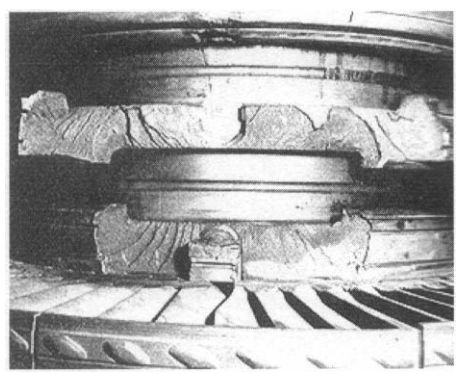


Figure 1.10 Failed turbine disc

(f) Airline propeller failure

Figure 1.11 shows a failed airscrew. These engineering components are exposed to very high time-dependent loads. Consequently, the most common mode of failure of propellers is fatigue.

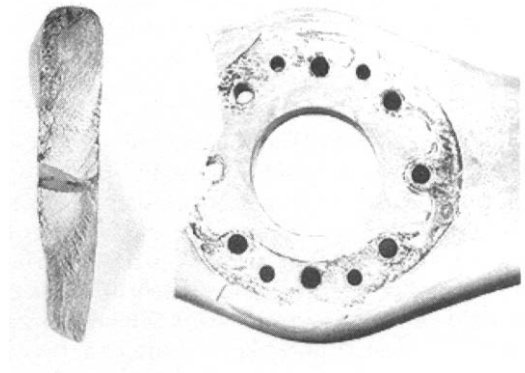


Figure 1.11 Failed aircraft propeller

In Figure 1.11, the varied discolouration on the failed section clearly indicates this type of failure. Also, the failure commenced at one of the bolt holes on the propeller, indicating that this was a region of high stress caused by the bolt holes acting as stress concentrators.

As can be expected, propeller failure will result in spectacular and occasionally tragic consequences. As a conservative safety measure, regulations require commercial aircraft to be able to fly and land with half of their normal complement of engines still operating. However, depending on the specific nature of the failure, corrective action by aircrew is not always possible.

Two cases were reported by Schaden Spiegel 1/98, in which twin engine turboprop aircraft, from the same manufacturer, both lost propellers in flight. One aircraft lost a propeller as it was about to land and eventually landed safely, incurring only mechanical damage. The other aircraft lost its propeller in mid flight, and had to make an emergency landing. Several people including the pilot were killed and the aircraft was lost. The resulting damage loss from this one incident was estimated at in excess of $US 100 million.

In both cases the failures were traced to a small surface indentation incurred during manufacture, resulting in a local stress concentration in an already highly stressed region near the root of the

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blade. Given the resulting high local stress, the resulting fatigue failures were predictable.

1.1.3 Summarising our review of failures

Almost all important failure cases are followed by an inquiry about the causes leading to the failure. These investigations often focus on *'who is to blame'* and *'who will bear the cost'* of the failure. Our focus is considerably more constructive. We focus on technical issues associated with failures, in order to learn from them. We hope to discover the underlying technical issues that might influence a whole class of such failures, and how to prevent this type of failure from happening again. There is an instructive story about a professional golfer, who was playing with a well known celebrity partner in a pro-celebrity golf tournament. The celebrity hit off at the first tee and landed his ball in the local car park, out of bounds. *"How the hell did I do that?"* cursed the celebrity. *"Why do you ask ?"*, questioned his professional partner, *"do you want to do it again?"*.

We have examined a series of ten failures. At the beginning of this chapter we categorised failures as *technical*, *operational* or *unpredictable*. In addition, we have identified a notion of *failure intensity* as a measure of economic, environmental and social impact caused by the failure. We should finally comment briefly on the problem of assessing *likelihood* of predictable failures.

In general, designers are aware of *normal*, or *expected*, or *most likely*, operating conditions of their engineering component or system, so that they are at least able to design for its *normal* operation. The difficulty arises when they have to plan for the *abnormal*. Ultimately, one could conceive of a situations so abnormal that it would be a *reasonable bet* that they would never arise. The implication is that design judgement is a little like gambling. Most engineers, however, are conservative, and wish to avoid the risk of gambling with financial or human loss. The most important judgement to be made early in any design, is the *worst credible accident* that is likely to befall our component or system. In many industries, codes of practice will prescribe allowable working stresses. In structural design, many codes also assign working loads. In mechanical design of engineering components, the assessment of working loads during the *worst credible accident* requires considerable experience and engineering wisdom.

Ex 1.1

(a) Categorise each of the ten failures described, as either *technical*, *operational*, or *unpredictable*.

(b) Rank order the ten failures described according to *failure intensity*.

Ex 1.2

We have identified economic, environmental, and social impact, as the three general results of failures. Are these effects independent? For example, can you identify a failure, not necessarily one of the ten listed, which might result in substantial environmental impact, without incurring proportionate economic impact?

Clearly, in most failures, the reverse is true: large economic impact with relatively minimal environmental impact.

Ex 1.3

Most of us face failure of engineering components in our daily lives. A washing machine or dishwasher fails in service, a component of our bicycle, motor car, or lawn mower, breaks, or the food processor, juicer, refrigerator, no longer perform as expected. Please identify the failure of some engineering component with which you have had direct contact. Describe the failure in terms of structural integrity as far as possible. Classify the most probable causes of failure. Use the failure categories offered above. For your component, identify the *worst credible accident* the designer should consider when designing it.

Ex 1.4

What is the *worst credible accident* for a door handle, a door hinge, the legs of an office chair, the top of an office desk, a power pole, a bar stool, a pair of scissors.

1.2 Units, estimation and things we expect you to know

In every domain of knowledge one needs to establish a frame of reference for effective communication. An amusing, albeit instructive, story is about a visit by the physicist Leo Szilard to the bacteriology laboratory of Salvador Luria, who was awarded the Nobel prize for his work on bacteriophages (viruses that attack bacteria). Szilard

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himself was by then an established figure in physics, partly responsible for the Manhattan project that developed the first atom bomb. Luria was initially embarrassed at showing the famous physicist through his laboratory. *"Dr. Szilard"* he asked, *"what should I assume, how much should I explain?" "You may assume"* responded Szilard, *"infinite ignorance and unlimited intelligence"* (Weber and Mendosa, 1973). In what follows we apply Szilard's mildly arrogant measure to readers of this text.

1.2.1 Units[1.4]

Whenever we are concerned with the discussion of physical quantities, for consistency and transportability of information the basic units of measure associated with each physical quantity need to be established. In this text we make exclusive use of the International System of Units (Système International d'Unités) commonly known as SI units.

The origin of SI units, based on the decimal system, dates to the aftermath of the French revolution, when in 1791 the French Academy of Sciences adopted the *metre* as the unit of length. This measure was based on 10^{-7} times the quadrant of a great circle of the Earth (meridian) passing through the poles and through Paris. It took six years to survey the arc from Barcelona to Dunkirk, resulting in a value of 39.37008 inches to the metre.

The units of *gram*, 10^{-6} times the mass of a m^3 of water at its maximum density (4°C), and *litre*, the volume of 10^{-3} m^3 of water, are both derived from the metre.

In 1875 a National Bureau of Weights and Measures was established at Sèvres, near Paris, where representatives of 40 countries meet every six years to update the international units of measurement.

For some time, the units of *metre* and *kilogram* were based on two convenient archival standards, known as *étalons*. For the metre, this was the distance between two marks on a specified bar of platinum, and for the kilogram, a carefully constructed and certified mass of platinum. At the 1983 meeting of the National Bureau of Standards, seven basic physical quantities were defined in the following way:

metre (m) = distance travelled by light in vacuum in 1/299,792,458 seconds;

kilogram (kg) = the mass of the internationally recognised archival prototype (*étalon*) at Sèvres;

second (s) = duration of 9,192,631,770 periods of the radiation corresponding to the transition between two hyperfine levels of caesium 133;

ampere (A) = the constant current which, when maintained in two straight parallel conductors 1 metre apart, generates a force of 10^{-7} newton per metre of length;

Kelvin (K) = 1/273.16 the absolute temperature of the triple-point of water (where the three phases are in equilibrium);

candela (luminous intensity) = 1/600,000 of the intensity of one m^2 of a radiating cavity at the temperature of freezing platinum (approximately the luminous intensity of one paraffin candle);

The more commonly found measure of luminous intensity is the unit of *lumen*. This is the quantity of visible light emitted per unit time per unit solid angle or steradian (recall that there are 4π steradians in a sphere). The lumen is related to the visual sensation caused by light and is dependent on the wavelength used. For example, light at a wavelength of 555 nanometres (10^{-9} metre), to which the eye is most sensitive, has a luminous intensity of 685 lumens per watt of radiant power. 1 lumen/steradian = 1 candela.

mole = the quantity of elementary substance equal to the number of atoms contained in 0.012 kilograms of carbon-12.

While we make exclusive use of SI units throughout this text, some engineering organizations still make use of the older Imperial System of units. It is expected that within the foreseeable future engineers will need to be familiar with both sets of units. A table of conversion factors is provided in the Appendix.

Ex 1.5

Convert between the following units :

(a) Fresh water has a density of 1000 $kg.m^{-3}$ (at 20°C). Express this in lbm ft^{-3} (note: 1 kg = 2.205 lbm; 1 m = 3.28 ft).

(b) Convert the thermal conductivity of copper from Imperial units to SI units (i.e. 225 Btu/hr-ft-°F to kW/m °K) (1 Btu = 1.055 kJ; 1W = 1J/s).

1.4 Source for this brief discourse on the origin of units of measurement is The Encyclopaedia Britannica, 15th edition.

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1.2.2 Estimation

Estimation is a euphemism for *informed guess*. Engineering science eschews guessing or *estimating* by the very nature of its analytical content. Engineering design, however, relies on estimation for synthesis. We cannot sensibly evaluate the outcomes of all possible choices in synthesising a solution to an engineering design problem. Hence, we often rely on estimation as a *heuristic* (*rule of thumb*), for reducing the task of evaluating outcomes to only a few, informed, solution choices. Naturally, there is a risk involved in making guesses, or estimates. But then engineering design synthesis is a risky business, not for the fainthearted.

There are many industries relying on estimation for their business, and accepting risk as an integral part of that business. Perhaps the most well recognised of these risky businesses is insurance. If we wish to be insured against some loss, we estimate the value of the loss, and contract with an insurance underwriter to *carry the loss* should it actually occur. The insurer estimates the nature and extent of the risk involved in the loss event, and based on this estimation, calculates an appropriate *premium*, or annual payment, to cover the payout when, and if, the loss event is incurred. In essence, the whole process is a gamble, where the insured bets with the insurer that the loss will occur, and continues to lay an annual wager to this effect. Since, generally, the wager is only a small proportion of the eventual loss value, both parties, the insurer and insured, benefit from the process.

In the business of insuring against industrial loss, a key parameter of insurance risk is experience with the technology involved. Insurers go to inordinate length in estimating risk with new technologies. The industrial insurance of space technology involved an almost unacceptable risk at the beginning of the 1970s. Not only was the technology new, with no data to rely on for loss estimation, but also the losses that might be incurred could be very large. A single space launch, not including the huge expense involved in development, can cost between $US 500 and 700 million (based on 1992 $US).

One would be tempted to question the value of offering industrial insurance in such a risky technological environment. However, the benefits to be gained are also substantial. At the end of 1992 there were some 300, mostly geo-stationary, satellites in orbit around the Earth. As experience with the technology grows, the insurance estimation becomes less risky. Figure 1.12 is a view of the space shuttle, the current pinnacle of technological achievement and complexity in the space industry. Also shown in Figure 1.12 is the insurance loss and gain data for the years 1976 to 1992. Clearly, while the insurers made substantial early losses, the industry appears to be breaking even cumulatively. In future years the space insurance industry can look forward to strong gains from the experience built up in these early startup years.

As already noted, engineering judgement is a key determinant of design success. The underlying skill in engineering judgement is the capacity to marshal

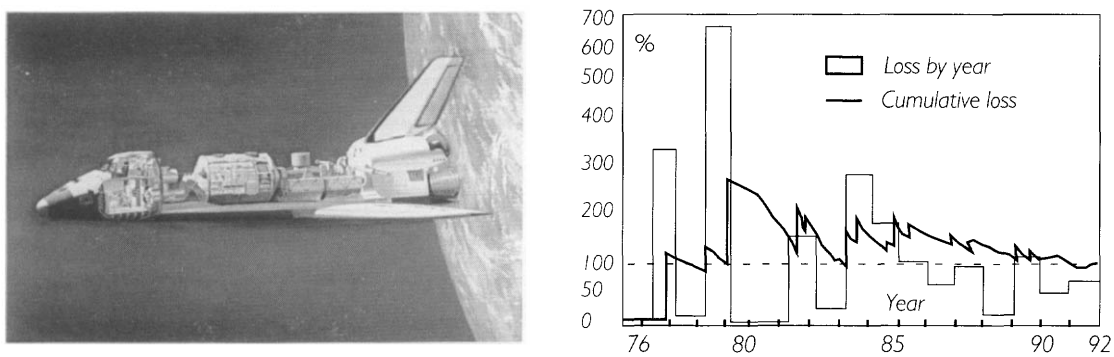


Figure 1.12 European space lab in cargo bay of the space shuttle and space insurance loss data: After Schaden Spiegel, 'Space Flight and Insurance' 1993,

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all the available data and to make informed, intelligent estimates about loads and failure scenarios. Typically, we can find almost limitless examples where quite realistic population estimates may be drawn from personal experience and observation.

An example will demonstrate the process of estimation. Suppose we were interested in designing hospital equipment to support birthing in hospital maternity wards. As a matter of course, we would need to estimate the birthrate to asses the size of our market. Figure 1.13 provides the necessary data for our estimate.

Australian population $\approx$ 18 million

Victorian population $\approx 0.25 \times 18$ million $= 4.5$ million

$\Rightarrow$ approximately $4.5/2 = 2.25$ million women.

Figure 1.13 indicates the average life expectancy of Australian females is 75 years and the birthrate has stabilised at around 2 children per female. This means that every female will, on average, give birth to 2 children in a lifetime of 75 years.

We can now estimate the total number of children born as follows :

$$N_{births} = 2.25 \times 10^6 \times (2/75) = 60{,}00 \text{ births.}$$

Bureau of Statistics data shows that in 1986 there were 60,650 live births to a Victorian population of 4.25 million.

Clearly, we were able to make a close estimate of birthrate from the demographic data of Figure 1.13, as well as our own wisdom about the total population of Australia, and the proportion of population living in the State of Victoria. Engineering estimation proceeds in a way not very different from the above example. Of course, for success in estimation and engineering judgement, we need to start with a list of what we expect you to know.

1.2.3 What we expect you to know

We begin with a brief, albeit random, list of *general knowledge* information. These are listed in Table 1.1 as a series of questions and we encourage first time readers to test themselves, by covering the answer column, and attempting to answer the questions posed. Where calculations are needed, these should be done mentally. Total time estimate for answering all these questions successfully is 10 minutes.

While we focus here on factual knowledge, it is useful to recall the admonishment of Henri Poincarré (1854-1912) :

"Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more science than a heap of stones is a house."

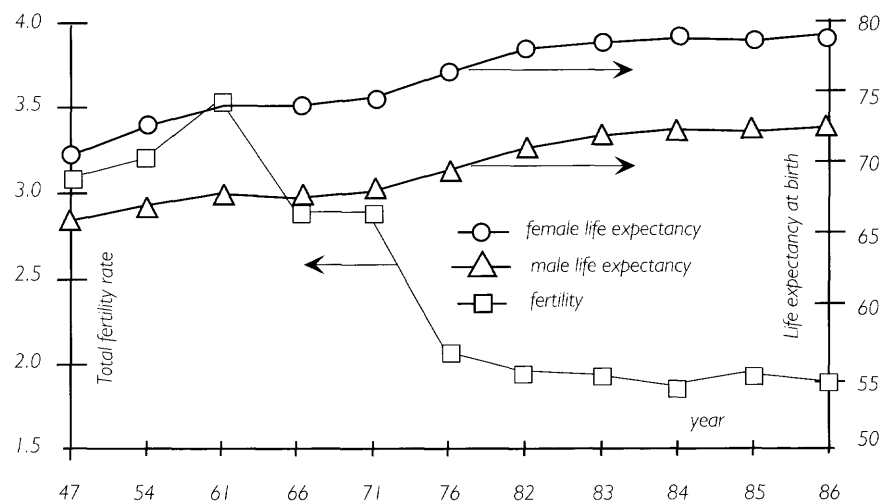


Figure 1.13 Australian fertility and mortality data
(Source: Australian Bureau of Statistics Cat. No. 3101.0, Australian Demographic Statistics)

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Table 1.1 General knowledge quiz

What is the speed of light in vacuum (in $m.s^{-1}$) – 2 significant figures ?	3.0×10^8
What is the 2nd moment of area of a rectangular section of width b and height d about a horizontal centroidal axis ?	$bd^3/12$
What is the 2nd moment of area of a circular section of diameter D about a transverse centroidal axis ?	$\pi D^4/64$
What is the density of steel (in $kg.m^{-3}$) ?	7,800
What is the density of aluminium (in $kg.m^{-3}$) ?	2,700
What is the speed of sound in air (in $m.s^{-1}$) ?	300
What is the value of Young's Modulus for steel ?	210 GPa
What is the value of Young's Modulus for aluminium ?	70 GPa
What does the unit prefix *giga* indicate ?	10^9
What does the unit prefix *pico* indicate ?	10^{-12}
What is the diameter of the earth ?	12,756 km (equatorial)
What is the area of a sphere, as a function of its radius ?	$4\pi R^2$
What is the volume of a sphere, as a function of its radius ?	$4\pi R^3/3$
What is the maximum bending moment in a simply supported beam of length L when loaded by a central concentrated force P ?	PL/4
What is the maximum bending moment in a simply supported beam of length L when loaded by a total force P distributed uniformly along its length ?	$PL/8$
What is the shape of the BMD for a cantilever subject to a uniformly distributed load ?	parabolic
What is the expression for the bending stress in a beam having a 2nd moment of area I_{zz}, subject to a bending moment of M ?	$\sigma = My/I_{zz}$ (y=distance from neutral axis)
What is the rotational analogue of the relationship: $F = m(d^2x/dt^2)$?	$T = J(d^2\theta/dt^2)$
What is the yield stress of mild steel ?	240 MPa
What is the current human population of the earth (to one sig. fig.) ?	6×10^9

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However, attempting to work on problems of significance in engineering design, without the background knowledge of basic facts, is like trying to build a house without foundations.

One of the oldest cliché (*hackneyed, overused phrase*) in engineering is the *back of the envelope* calculation. No doubt, this phrase predates computers and electronic calculators. It was often used by practising engineers as a pseudonym for *estimation involving a few facts and numbers*. In essence, the above estimation of birthrates is an example of a *back of the envelope* calculation.

(a) An estimation quiz

The following quiz is to do with estimation. We don't need to calculate any of the results in this quiz, or, at least, if there is to be calculation, it should be of the *back of the envelope* type. In this second quiz, you should try to answer all the questions in 60 seconds.

(a) What mass can the average person lift and carry comfortably over 100 metres ?

(b) What is the height of a bull elephant ?

(c) What is the mass of a bull ant ?

(d) What is the weight of a moderate-sized apple ?

(e) What is the maximum squeezing force which could be delivered to a pair of pliers from the average hand ?

(f) How many locusts in a locust swarm ?

Approximate answers: (a) 30kg; (b) 3m; (c) 0.1gm; (d) 1N; (e) 200N; (f) 10^9.

(b) Calculator quiz

The next quiz allows the use of calculators, and in this case the aim is for accuracy rather than quick results. Correspondingly more care is needed in estimating answers than in the previous two quizzes.

(a) What is the contact pressure under a stiletto heel ?

(b) What is it under the foot of an elephant ? (average weight 5 tonnes)

(c) How many cars cross Golden Gate Bridge in San Francisco each day ?

[Hint: The population of San Francisco is approximately 800,000. We can guess that the Golden Gate Bridge, 1280m midspan, is the major thoroughfare for crossing the harbour and about $1/16^{th}$ of the population cross it daily: is this estimate reasonable ?]

(d) What is the average number of cars on the Golden Gate Bridge at one time ?

[Hint: Consider the length of time taken to cross the midspan as 4 minutes: is this reasonable ?]

(e) What is the average daily electrical energy consumption in California (in GJ/day) ?

[Hint: The power consumption of the USA is approximately 3 x 10^{12}kW-hour per annum, and the population of USA is 260 million and that of California 30 million]

(f) How much water is flushed down London toilets in one day ?

[Hint: The population of Greater London is approximately 7 million]

Approximate answers : (a) 10MPa; (b) 0.17MPa; (c) 50,000; (d) 280; (e) 3.4×10^6 GJ; (f) 20×10^6 litres.

(c) Mental calculations

The final quiz involves mental calculation only.

(a) How many seconds in a year ?

(b) Evaluate $\sqrt{800}$.

(c) The Empire State Building has 102 floors and a TV mast above ground. How high is it ?

(d) What volume of water is contained in an Olympic swimming pool ?

(e) How many Monday afternoons have there been since 1 A.D. ?

Answers : (a) 32 million; (b) 449 metres; (c) $20\sqrt{2}$; (d) 2,000 m^3; (e) 104,000.

We place considerable importance on the capacity to make rational estimations in engineering. This has become particularly important since the widespread use of electronic calculators. It is so easy to *slip a digit or two* when entering data in a complicated set of calculations. In general, one needs to invoke self-checking mechanisms to ensure realistic results are obtained for problems where there is little or no previous experience on the expected order of magnitude of

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the results. Order of magnitude, and dimensional consistency, checks must be routinely used for all calculations. Moreover, the number of significant figures available from electronic calculators bears no relation to the real precision of our results. Consequently, we never present more significant figures in our results than we can credibly justify, within the levels of approximations made along the way. Ultimately, these procedures build confidence and credibility in evaluations for both engineer and client. Our guiding principle must be that we engineers are *in the credibility business*.

(d) Rational estimations:

- focus on the essential issues behind an unfamiliar problem;
- help to gain insight into the way key controlling parameters influence a process or outcome;
- help to concentrate on primary rather than second-order effects (*see the wood for the trees*);
- help to identify major sources of uncertainty.

Ex 1.6

Calculate the number of barbers that could be fully employed within the state of California [population 30 million] (see solution at 1.5.1).

Ex 1.7

Check the following *dimensionless* numbers for dimensional consistency

(a) Reynolds Number $Re = \dfrac{\rho V D}{\mu}$

(b) Nusselt Number $Nu = \dfrac{h_x x}{k}$

(c) Grashof Number $Gr = \dfrac{g\beta\left(T_w - T_\infty\right)x^3}{\nu^2}$

[ρ = density; V = velocity; D = characteristic length; μ = dynamic viscosity; h_x = convective heat transfer coefficient at a distance x from initial contact with the surface; k = thermal conductivity; g = gravitational acceleration; β = volumetric coefficient of thermal expansion; ν = kinematic viscosity = μ / ρ, with units of $m^2.s^{-1}$]

Ex 1.8

The following formula gives the wind *drag* per unit length on a cylindrical body of diameter D exposed to a wind of velocity V [C_D is a dimensionless *drag coefficient*].

$$q = C_D\left(\frac{\rho V^2}{2}\right)D.$$

Check this formula for dimensional consistency.

(e) Note on the methodical conversion of units:

- arrange everything to be clearly in the numerator or the denominator;
- use the *brackets operator*, [...], to indicate *"the units of..."*;
- place units (in parentheses) alongside each simple quantity;
- convert each simple quantity one-at-a-time;
- cancel and calculate.

Example 1: Convert a flow rate of 1000 UK gal/hr to ℓ/s

$$Q = 1000\left(\frac{\cancel{\text{UK gall}}}{\cancel{\text{hr}}}\right)\times 4.546\left(\frac{\ell}{\cancel{\text{UK gal}}}\right)\times\frac{1}{3600}\left(\frac{\cancel{\text{hr}}}{\text{s}}\right)$$

$$= \frac{1000\times 4.546}{3600}\left(\frac{\ell}{\text{s}}\right) = 1.26\ \ell/\text{s}$$

Example 2: Check the units of the universal gravitational constant in

$$F = \frac{GMm}{r^2}.$$

$$\left[G\right] = \left[\frac{Fr^2}{Mm}\right] = \frac{\left[F\right]\left[r^2\right]}{\left[M\right]\left[m\right]} = \frac{\text{N.m}^2}{\left(\text{kg}\right)^2} = \frac{\cancel{\text{kg}}.\text{m}.\text{m}^2}{\text{s}^2\left(\cancel{\text{kg}}\right)^2}$$

$$= \text{m}^3\,\text{s}^{-2}\,\text{kg}^{-1}$$

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(f) Key elements of clear engineering calculations:

- the work is well spaced;
- assumptions and simplifications are stated unambiguously;
- the argument unfolds in a logical sequence;
- there is a good balance between algebra and calculation;
- useful intermediate values are calculated;
- important intermediate results are highlighted;
- related results are summarised (maybe tabulated);
- you *know* when you've got a ridiculous answer;
- awareness of *meaningful accuracy* (significant figures).

Ex 1.9 'Chessboard, wheat, and a constipated sparrow'

According to legend, the inventor of the game of chess asked from his grateful mogul lord only one thing in payment — all the wheat which would be needed on a 8x8 chess board if one were to place 1 grain on the first square, 2 grains on the second, 4 on the third, and so on, doubling up on each subsequent square.
Imagine that all this wheat is eaten by a sparrow. Calculate the diameter of such a sparrow, assuming the bird does not excrete any wastes until it finishes its meal. (see solution at 1.5.2)
(It is a popular belief that the inventor was executed for trying to be a bit too clever)

Ex 1.10 'Energy from a cuppa'

An urban legend has it that James Prescott Joule (1818-1889), the English physicist after whom the unit of energy is named, took several thermometers with him on his honeymoon to Niagara falls.
(a) Estimate the temperature rise due to a conversion of potential energy into heat over the 49 metre falls;
(b) Calculate the height to which a cup of tea could be lifted above sea level, if all the thermal energy in the hot fluid were converted into mechanical energy. It is an interesting *party trick* to try to estimate the height initially without calculating.
(see solution at 1.5.3)

CASE EXAMPLE 1.1: DOLPHIN ENCLOSURE

A cautionary tale about engineering calculations
Source: *Checking designs is equally important as designing itself,* James Ferry, *Engineers Australia*, Feb., 1995: 52–53

"... The case concerned the design and construction of a relatively simple and inexpensive dolphin enclosure."

"... A crucial calculation, using an equation known as Merion's equation, was made at conceptual design stage."

"... The calculation contained a simple and easily identifiable arithmetical error. However, it was never checked."

"... almost immediately after the vinyl enclosure was constructed, [it] failed as a result of the presence of long waves in the marina."

"... [the client] ultimately claimed ... the losses it had suffered as a result of the defective design. These losses, inclusive of interest, by the end of the hearing totalled $602,845.68, together with legal and expert witness costs of some $150,000.00." [The original firm of designers and the company responsible for checking were both sued. The judge held them both to be equally liable for the failure.]

"... The case illustrates the importance of risk management systems where all designs are reviewed and checked."

CASE EXAMPLE 1.2: A CHRISTMAS TALE

"Is there a Santa Claus?"
This question is considered in terms of *realistic* estimates of what Santa needs to do in order to deliver the *goods* on Christmas eve.

1. There are two billion children (persons under 18) in the world. Since roughly 15% of the world's population is Christian, it is at least possible that there are 300 million children who believe in Santa Claus, and who therefore qualify for a visit (let us assume that they are all *good*). At an average of 3.5 children per household there are 86 million households.
2. Allowing for time zones and the rotation of the Earth, assuming he travels east to west, Santa has 31 hours of Christmas to work with. This represents 770.6 visits per second.

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